

CHAPTER 2

PROBABILITY

2.1 INTRODUCTION

The concept of probability has been an object of debate among philosophers, logicians, mathematicians, statisticians, physicists, and psychologists for the last couple of centuries, and this debate is not likely to be over in the foreseeable future. As advocated by Bertrand Russell in his essay on skepticism, when experts disagree, the layman would do best by refraining from forming a strong opinion. Accordingly, we will not enter into the discussion about the nature of probability; rather, we will start from the issues and principles that are commonly agreed upon.

Probability is a number associated with an event that is intended to represent its “likelihood,” “chance of occurring,” “degree of certainty,” and so on. Probabilities can be obtained in several ways, the most common being (1) the *frequency* (or *objective*) interpretation, (2) the *classical* (sometimes called *logical*) interpretation, and (3) the *subjective* or *personal* interpretation of probability.

2.2 PROBABILITY AS A FREQUENCY

According to the common interpretation, probability is the “long-run” relative frequency of an event. The idea connecting probability and frequency is (and had been for a long time) very well grounded in everyday intuition. For instance, loaded dice were on several occasions found in the graves of ancient Romans. That indicates that they were aware of the possibility of modifying long-run frequencies of outcomes, and perhaps making some profit in such a way.

Today, the intuition regarding relationship between probabilities and frequencies is even more firmly established. For instance, the phrases “there is 3% chance that an orange picked

at random from this shipment will be rotten” and “the fraction of rotten oranges in this shipment is a 3%” appear almost synonymous. But on closer reflection, one realizes that the first phrase refers to the probability of an event “randomly selected orange will be rotten,” while the second phrase refers to the population of oranges.

The precise nature of the relation between probability and frequency is hard to formulate. But the usual explanation is as follows: Consider an experiment that can be repeated under identical conditions, potentially an infinite number of times. In each of these repetitions, some event, say A , may occur or not. Let $N(A)$ be the number of occurrences of A in the first N repetitions. The frequency principle states that the ratio $N(A)/N$ approximates the probability $P(A)$ of event A , with the accuracy of the approximation increasing as N increases.

Let us observe that this principle serves as a basis for estimating probabilities of various events in the real world, especially those probabilities that might not be attainable by any other means (e.g., the probability of heads in tossing a biased coin).

We start this chapter by putting a formal framework (axiom system) on a probability regarded as a *function on the class of all events*. That is, we impose some general conditions on a set of individual probabilities. This axiom system, due to Kolmogorov (1933), will be followed by the derivation of some of its immediate consequences. The latter will allow us to compute probabilities of some composite events given the probabilities of some other (“simpler”) events.

2.3 AXIOMS OF PROBABILITY

Let S be the sample space, namely the set of all outcomes of an experiment. Formally, probability, to be denoted by P , is a function defined on the class of all events,¹ satisfying the following conditions (usually referred to as *axioms*):

Axiom 1 (Nonnegativity):

$$P(A) \geq 0 \text{ for every event } A.$$

Axiom 2 (Norming):

$$P(S) = 1.$$

Axiom 3 (Countable Additivity):

$$P(A_1 \cup A_2 \cup \cdots) = \sum_{i=1}^{\infty} P(A_i)$$

for every sequence of pairwise disjoint events $A_1, A_2, \dots$, (such that $A_i \cap A_j = \emptyset$ for all $i \neq j$). $\square$

If the sample space S is finite or countable, one can define a probability function P as follows: Let f be a nonnegative function defined on S , satisfying the condition $\sum_{s \in S} f(s) = 1$. Then, P may be defined for every subset A of S as $P(A) = \sum_{s \in A} f(s)$. One can easily check that P satisfies all three axioms.

Indeed, $P(A) \geq 0$ because f is nonnegative, and $P(S) = \sum_{s \in S} f(s) = 1$. Finally, let $A_1, A_2, \dots$ be a sequence of disjoint subsets of S . Then,

$$\begin{aligned} P(A_1) + P(A_2) + \cdots &= \sum_{s \in A_1} f(s) + \sum_{s \in A_2} f(s) + \cdots \\ &= \sum_{s \in A_1 \cup A_2 \cup \cdots} f(s) = P\left(\bigcup_{i=1}^{\infty} A_i\right). \end{aligned}$$

¹The nature of the class of all events will be (to a certain extent) explicated in Section 2.6. See also Section 1.4.

Passing from the first to the second line is allowed because $A_1, A_2, \dots$ are disjoint, so each term appears only once. The sum in the second line is well defined (i.e., it does not depend on the order of summation because the terms are nonnegative).

However, if S is not countable, one usually needs to replace summation by integration, $P(A) = \int_A f(s) ds$. This imposes some conditions on functions f and on the class of events A . For a detailed discussion, the reader is referred to more advanced probability texts (e.g., Chung, 2001).

■ EXAMPLE 2.1 Geometric Probability

One of the first examples of an uncountable sample space is associated with “the random choice of a point from a set.” This phrase is usually taken to mean the following: a point is selected at random from a certain set S in a finite-dimensional space (line, plane, etc.), where S has finite measure (length, area, etc.), denoted generally by $|S|$. The choice is such that if A ($A \subset S$) has measure $|A|$, then the probability of the chosen point falling into A is proportional to $|A|$. Identifying S with the sample space, we can then write $P(A) = |A|/|S|$.

To better see this, suppose that in shooting at a circular target S , one is certain to score a hit, and that the point where one hits S is assumed to be chosen at random in the way described above. What is the probability that the point of hit is farther from the center than half of the radius of the target?

From Figure 2.1, it is clear that the point of hit must lie somewhere in the shaded annulus A . Its area is $|A| = \pi R^2 - \pi(R/2)^2$ so that $P(A) = |A|/\pi R^2 = 3/4$. Of course, the assumption under which this solution is obtained is not very realistic: typically sets closer to the center are more likely to be hit than sets of the same area located closer to the perimeter.

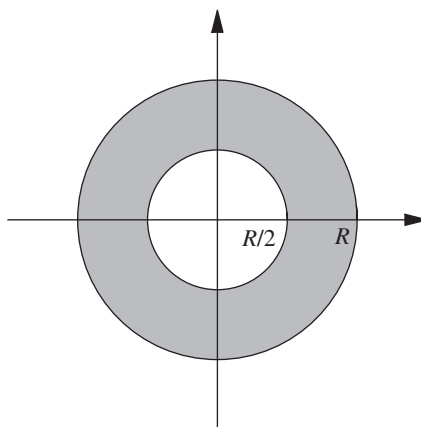


Figure 2.1 Hitting a target.

The concept of “random choice” from an uncountable set is sometimes ambiguous. This is illustrated by the next example.

■ EXAMPLE 2.2 Bertrand’s Paradox

A chord is chosen at random in a circle. What is the probability that the length of the chord will exceed the length of the side of an equilateral triangle inscribed in the circle?

This problem was originally posed in 1888 by Joseph Bertrand, a French mathematician, who provided three solutions, all valid, but yielding inconsistent results.

SOLUTION 1. The chord is uniquely determined by the angle α (see Figure 2.2). These angles are chosen at random from the interval $(0, \pi)$. It is clear that the length of the chord exceeds the side of the equilateral triangle if α lies between $\pi/3$ and $2\pi/3$, so the answer to the question is $1/3$.

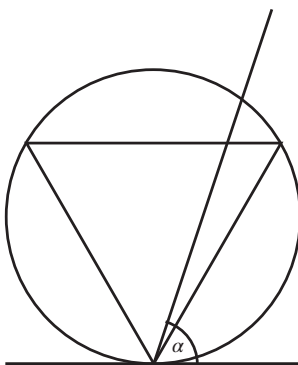


Figure 2.2 First solution of Bertrand's problem.

SOLUTION 2. Let us draw a diameter QQ' (see Figure 2.3) perpendicular to the chord P . Then, the length of the chord exceeds the side of the equilateral triangle if it intersects the line QQ' between points B and B' . Elementary calculations give $|BB'| = |QQ'|/2$, so the answer is $1/2$.

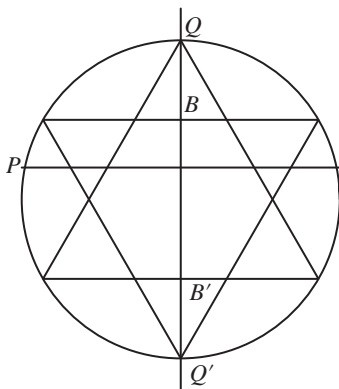


Figure 2.3 Second solution of Bertrand's problem.

SOLUTION 3. The location of the chord is uniquely determined by the location of its center (except when the center coincides with the center of the circle, which is an event with probability zero). For the chord to be longer than the side of the equilateral triangle inscribed in the circle, its center must fall somewhere inside the shaded circle in Figure 2.4. Again, by elementary calculations, we obtain probability $1/4$.

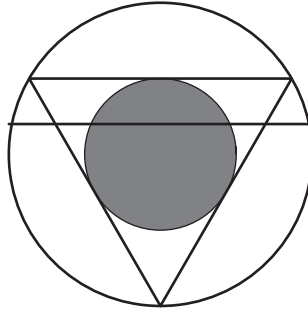


Figure 2.4 Third solution of Bertrand's problem.

The discovery of Bertrand's paradox was one of the impulses that made researchers in probability and statistics acutely aware of the need to clarify the foundations of the theory, and ultimately led to the publication of Kolmogorov's book (1933). In the particular instance of the Bertrand "paradoxes," they are explained simply by the fact that each of the solutions refers to a different sampling scheme: (1) choosing a point on the circumference and then choosing the angle between the chord and the tangent at the point selected, (2) choosing a diameter perpendicular to the chord and then selecting the point of intersection of the chord with this diameter, and (3) choosing a center of the chord. Random choice according to one of these schemes is not equivalent to a random choice according to the other two schemes.

To see why it is so, we will show that the first and second scheme are not equivalent. The analogous arguments for the other two possible pairs of schemes are left as an exercise.

■ EXAMPLE 2.3

Imagine different devices (physical mechanisms, computer programs, etc.) built for sampling random chords. One scheme chooses a point on the circumference, and then the angle α between the chord and the tangent to the circle at the point chosen (Figure 2.2). The second scheme chooses first the direction of the diameter and then the point B on the diameter, at which the chord perpendicular to this diameter intersects it (Figure 2.3). From Figure 2.5, it is seen that the angle AOB is α , and

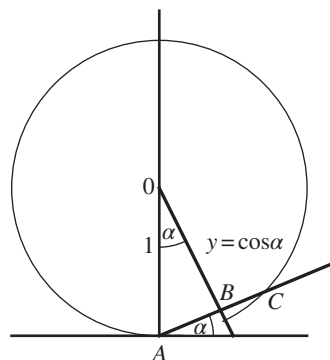


Figure 2.5 Explanation of Bertrand's paradox.

therefore $y = |OB| = \cos \alpha$. Thus, $dy = (\sin \alpha) d\alpha$, which means that equal changes of α do not produce equal changes of y . In fact, these changes are smaller when α is small. Consequently, a device that chooses angles α at random will tend to produce more intersections of the diameter that are farther from the center (i.e., more chords will be shorter).

PROBLEMS

- 2.3.1** Label all statements below as true or false. **(i)** If A is more likely to occur than A^c , then $P(A) > 0.5$. **(ii)** If A occurs whenever B does, then $P(B) \leq P(A)$. **(iii)** If $P(A) \leq P(B)$, then whenever B occurs, A does also. **(iv)** If $P(A) = 0.75$, then A must occur three times out of every four. **(v)** The sum of probabilities of disjoint events A and B cannot exceed 1. **(vi)** If A and B are not disjoint, the sum of their probabilities exceeds 1. **(vii)** If $P(A \cap B)$, $P(A \cap C)$, and $P(B \cap C)$ are all positive, then $P(A \cap B \cap C)$ is also positive. **(viii)** If sample spaces for two experiments are identical, then the probability of the same event A must be the same for both experiments.
- 2.3.2** A bathroom floor is covered by square tiles with side length a . You drop a coin with diameter b , where $b < a$. Find: **(i)** The probability that the coin will rest entirely within one tile. **(ii)** The probability that the coin will partially cover four different tiles.
- 2.3.3** Show that first and third, as well as second and third, schemes of sampling chords (see Bertrand's paradox) are not equivalent.

2.4 CONSEQUENCES OF THE AXIOMS

The simplest consequences of the axioms of probability are as follows:

1. *The probability of the impossible event is zero:*

$$P(\emptyset) = 0. \quad (2.1)$$

This follows from the fact $1 = P(\mathcal{S}) = P(\mathcal{S} \cup \emptyset \cup \emptyset \cup \dots) = P(\mathcal{S}) + P(\emptyset) + P(\emptyset) + \dots$, which is possible only if $P(\emptyset) = 0$. It is important to realize that the converse is not true: the condition $P(A) = 0$ does not imply that $A = \emptyset$. This is shown by the following example:

■ EXAMPLE 2.4

Consider an experiment consisting of tossing a coin infinitely many times. The outcomes may be represented as infinite sequences of the form HHTHTTHT ... so that the sample space $\mathcal{S}$ contains infinitely many of such sequences. The event "heads only," that is, the set consisting of just one sequence HHHH ..., is not empty. However, the chance of such an outcome is, at least intuitively, zero: tails should come up sooner or later.

2. *Probability is finitely additive:*

$$P(A_1 \cup \dots \cup A_n) = P(A_1) + \dots + P(A_n)$$

for any $n = 1, 2, \dots$, if the events A_i are pairwise disjoint.

In an infinite sequence $A_1, \dots, A_n, \emptyset, \emptyset, \dots$, events are pairwise disjoint only if $A_1, \dots, A_n$ are, so Axiom 3 applies. Therefore, using (2.1), we have

$$\begin{aligned} P(A_1 \cup \dots \cup A_n \cup \emptyset \cup \dots) &= P(A_1) + \dots + P(A_n) + P(\emptyset) + \dots \\ &= P(A_1) + \dots + P(A_n), \end{aligned}$$

while the left-hand side is $P(A_1 \cup \dots \cup A_n)$.

3. *Monotonicity*: If $A \subset B$ then $P(A) \leq P(B)$. This follows from the fact that $B = A \cup (B \cap A^c)$. The events on the right-hand side are disjoint, so we have $P(B) = P(A) + P(B \cap A^c) \geq P(A)$ by Axiom 1. Since $A \subset \mathcal{S}$ for every event A , we have $0 \leq P(A) \leq 1$.

4. *Probability is countably subadditive*:

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) \leq P(A_1) + P(A_2) + \dots, \quad (2.2)$$

for every sequence of events $A_1, A_2, \dots$. This follows from representation (1.6) as a union of disjoint events, and then from monotonicity. We have

$$\begin{aligned} P\left(\bigcup_{n=1}^{\infty} A_n\right) &= P(A_1) + P(A_1^c \cap A_2) + P(A_1^c \cap A_2^c \cap A_3) + \dots \\ &\leq P(A_1) + P(A_2) + P(A_3) + \dots. \end{aligned}$$

5. *Complementation*:

$$P(A^c) = 1 - P(A). \quad (2.3)$$

This follows from Axiom 2, by the fact that A and A^c are disjoint and $A \cup A^c = \mathcal{S}$.

6. *Probability of a union of events*:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B). \quad (2.4)$$

Indeed, since $A \cup B = A \cup (A^c \cap B)$, then $P(A \cup B) = P(A) + P(A^c \cap B)$. On the other hand, $B = (A \cap B) \cup (A^c \cap B)$, and hence, $P(B) = P(A \cap B) + P(A^c \cap B)$. Solving for $P(A^c \cap B)$ in one equation and substituting into the other, we obtain (2.4).

A more intuitive argument may be made by using Venn diagrams (see Figure 2.6). In the sum $P(A) + P(B)$, each sample point from the intersection $A \cap B$ is included twice, so to obtain the probability of the union $A \cup B$, we must subtract the probability $P(A \cap B)$.

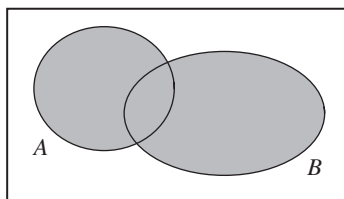


Figure 2.6 Union of two events.

■ EXAMPLE 2.5

Suppose that $P(A) = 0.4$, $P(B^c) = 0.6$, $P(A^c \cap B) = 0.1$. Find $P(A \cup B^c)$.

SOLUTION. The best strategy for solving this kind of problem is finding the probabilities of all the intersections (in this case, $A \cap B$, $A \cap B^c$, $A^c \cap B$, $A^c \cap B^c$).

The probability $P(A^c \cap B) = 0.1$ is given. Next, $(A \cap B) \cup (A^c \cap B) = B$, and the events on the left are disjoint. So $P(A \cap B) + P(A^c \cap B) = P(B) = 1 - P(B^c)$, which means that $P(A \cap B) + 0.1 = 1 - 0.6$, and hence, $P(A \cap B) = 0.3$. Then, using $A = (A \cap B) \cup (A \cap B^c)$, we have $0.4 = P(A) = P(A \cap B) + P(A \cap B^c) = 0.3 + P(A \cap B^c)$; hence, $P(A \cap B^c) = 0.1$. Similarly $P(A^c \cap B^c) = 0.5$. Applying formula (2.3), we have $P(A \cup B^c) = P(A) + P(B^c) - P(A \cap B^c) = 0.4 + 0.6 - 0.1 = 0.9$.

For the case of three events: A , B , and C , the same argument based on Venn diagrams gives the formula

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) \\ &\quad - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ &\quad + P(A \cap B \cap C). \end{aligned} \quad (2.5)$$

It can be checked at Figure 2.7 that the formula (2.5) includes each part of the union $A \cup B \cup C$ exactly once.

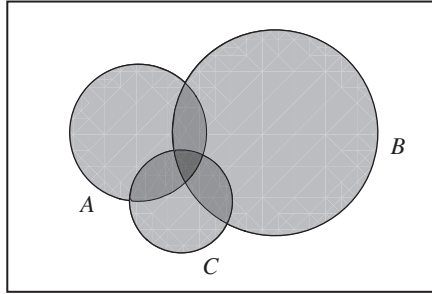


Figure 2.7 Union of three events.

Formula (2.5) may be generalized to the case of the union of any finite number of events.

Theorem 2.4.1 For any events $A_1, \dots, A_n$

$$\begin{aligned} P(A_1 \cup \dots \cup A_n) &= \sum_{i=1}^n P(A_i) - \sum_{1 \leq i_1 < i_2 \leq n} P(A_{i_1} \cap A_{i_2}) \\ &\quad + \sum_{1 \leq i_1 < i_2 < i_3 \leq n} P(A_{i_1} \cap A_{i_2} \cap A_{i_3}) + \dots \\ &\quad + (-1)^{n+1} P(A_1 \cap \dots \cap A_n). \end{aligned} \quad (2.6)$$

Proof: We will proceed by induction. The theorem is true for $n = 1$. Assume now that formula (2.6) holds, and write

$$P\left(\bigcup_1^{n+1} A_i\right) = P\left(\bigcup_1^n A_i\right) + P\left(A_{n+1} \setminus \bigcup_1^n A_i\right).$$

Since

$$P(A_{n+1}) = P\left(A_{n+1} \cap \bigcup_1^n A_i\right) + P\left(A_{n+1} \setminus \bigcup_1^n A_i\right),$$

we have

$$P\left(\bigcup_1^{n+1} A_i\right) = P\left(\bigcup_1^n A_i\right) + P(A_{n+1}) - P\left[\bigcup_1^n (A_i \cap A_{n+1})\right].$$

Applying formula (2.6) to $P(\bigcup_1^n A_i)$ and to $P[\bigcup_1^n (A_i \cap A_{n+1})]$, and then combining the corresponding terms, we obtain (2.6) with n replaced by $n + 1$. $\square$

PROBLEMS

- 2.4.1** Let events A , B , and C be such that A contains B and is disjoint from C . Moreover, A is twice as likely as B , three times as likely as C , and half as likely as its complement A^c . Find $P(B \cup C)$.
- 2.4.2** Events A , B , and C are such that $A \cap B = A \cap C = \emptyset$ while $B \cap C$ is not empty. Determine $P(A \cup B \cup C)$ as a function of x if $P(A) = P(B) = P(C) = 3P(B \cap C) = x$.
- 2.4.3** Find $P(A \cup B^c)$ for events A and B such that $P(A) = P(B) = 1/2$, and $P(A^c \cap B^c) = 1/3$.
- 2.4.4** To make the formula $P[(A^c \cap B^c)^c] = 2 - P(A^c) - P(B^c) - x$ valid, x must equal: (1) $P(A \cup B)$. (2) $P(A \cap B)$. (3) $P(A^c \cup B^c)$. (4) None of the above.
- 2.4.5** Three events A , B , and C are such that C is contained in the intersection $A^c \cap B$, $P(A) = 0.6$, $P(A \cap B) = 0.3$, and $P(C) = P(A^c \cap B^c) = 0.1$. Find: (i) $P(B)$. (ii) The probability that exactly two of the events A , B , C will occur. (iii) The probability that exactly one of the events A , B , C will occur.
- 2.4.6** Let A , B , C , D , and E be five events such that $P(A) = P(B) = \cdots = P(E) = k$ and $P(A \cap B) = P(A \cap C) = \cdots = P(D \cap E) = p$. Moreover, at least one of the events $A, \dots, E$ must occur, and the intersection of any three events among $A, B, \dots, E$ is empty. (i) Find p if $k = 0.3$. (ii) Omit the assumption that at least one of the events must occur and determine all possible values of k if it is known that $p = 0.01$.
- 2.4.7** If events $A_1, \dots, A_{101}$ are such that $P(A_1) = \cdots = P(A_{101}) = 0.01$, $P(A_1 \cap A_2) = P(A_1 \cap A_3) = \cdots = P(A_{100} \cap A_{101}) = r$, while every triple intersection is empty, find the smallest possible value of the probability of intersection r .
- 2.4.8** Before the era of cell phones (a time that most of today's students do not remember), people often used public phones that operated after a coin was inserted. These phones were sometimes faulty and either took a coin without giving a connection or gave the connection but also returned the coin. (i) A faulty public phone returns the coin with probability 60%, it gives you the number you dial with probability 20%, and it takes your coin and does not give you the required connection with probability 30%. Find the probability that you will talk with the number you dial for free. (ii) A certain public phone is such that it returns the coin with probability a , connects you with the number you dial with probability b , and it gives you the connection for free with probability c . Let us agree to say that the phone is *individually honest* if it takes your money if and only if it provides you with the required connection, and that it is *socially honest* if it takes, on average, as many coins as it gives correct connections (but perhaps from different customers). Find conditions for a , b , and c under which the phone is individually honest. (iii) Find conditions under which the phone is socially honest.

- 2.4.9** Based on a survey of undergraduate sophomores at a certain university, the researchers found out that 67% of students live in a dorm (D), 52% are interested in studying abroad (A), and 48% are planning to attend graduate school (G). Furthermore, 32% live in a dorm and are planning studying abroad, 30% are planning to study abroad and continue their education in a graduate school, while 30% live in a dorm and plan to apply to graduate schools. Moreover, 20% of sophomores answered all three questions positively ($D \cap A \cap G$). What is the probability that a randomly selected sophomore at this university either lives in a dorm, or is thinking of studying abroad, or is planning to apply to a graduate school?
- 2.4.10** A die is loaded in such a way that the probability of j dots on the top face is proportional to j , for $j = 1, 2, \dots, 6$. What is the probability that in one roll of the die an odd number of dots will turn up?

2.5 CLASSICAL PROBABILITY

For the so-called classical or logical interpretation of probability, we will assume that *the sample space S contains a finite number N of outcomes and all of these outcomes are equally probable*.

Obviously, in this case, each of the outcomes has the same probability $1/N$, and for every event A ,

$$P(A) = \frac{\text{number of outcomes in } A}{N}. \quad (2.7)$$

In many real situations, the outcomes in the sample space reveal a certain symmetry, derived from physical laws, from logical considerations, or simply from the sampling scheme used. In such cases, one can often *assume* that the outcomes are equiprobable and use (2.7) as a rule for computing probabilities. Obviously, the function P in (2.7) satisfies the axioms of probability.

To use some very simple examples, in tossing a regular die, each face has the same probability $1/6$. Then the probability of the event $A = \text{“outcome odd”}$ is $P(A) = 3/6 = 1/2$, since there are three odd outcomes among the possible six.

The case above is rather trivial, but considerations of symmetry can sometimes lead to unexpectedly simple solutions of various problems.

■ EXAMPLE 2.6

Peter tosses a fair coin n times, and Paul tosses it $n + 1$ times. What is the probability that Paul tosses more heads than Peter?

SOLUTION. Either Paul tosses more heads than Peter (event A) or he tosses more tails than Peter (event B). These two events exclude one another and exhaust all possibilities (since one cannot have ties in number of heads *and* number of tails). Switching the role of heads and tails transforms one of these events into the other. Thus, sample space becomes partitioned into two equiprobable events, and we must have $P(A) = 1/2$.

The use of (2.7) requires techniques for counting the numbers of elements in some sets. These topics, known under the name *combinatorics*, will be discussed in Chapter 3.

PROBLEMS

- 2.5.1** A coin is tossed seven times. Assume that each of the $2^7 = 128$ possible outcomes (sequences like HTTHHTH of length 7) is equally likely. Relate each outcome to a binary number by replacing H by 1 and T by 0, for example, THHHTTH is $0111001 = 57$. Find the probability that a number generated in this way lies between 64 and 95 (inclusive on both sides).
- 2.5.2** A number X is chosen at random from the series 4, 9, 14, 19, $\dots$, and another number Y is chosen from the series 1, 5, 9, 13, $\dots$. Each series has 100 terms. Find $P(X = Y)$.
- 2.5.3** A regular die and a die with 2, 3, 5, 6, 7, and 8 dots are tossed together, and the total number of dots is noted. What is the probability that the sum is greater than or equal to 10?
- 2.5.4** Use formula (2.6) to find the number of primes not exceeding 100. [*Hint*: Assume that you sample one of the numbers 1, 2, $\dots$, 100. Let A_i be the event “the number sampled is divisible by i .” Determine $p = P(A_2 \cup A_3 \cup A_5 \cup A_7)$. Then the answer to the problem is $100(1 - p) + 3$ (why?).]

2.6 NECESSITY OF THE AXIOMS*

Looking at Axiom 3, one may wonder why do we need it for the case of countable (and not just finite) sequences of events. Indeed, the necessity of all three axioms, with only finite additivity in Axiom 3, can be easily justified simply by using probability to represent the limiting relative frequency of occurrences of events. Recall the symbol $N(A)$ from Section 2.1 for the number of occurrences of the event A in the first N experiments. The nonnegativity axiom is simply a reflection of the fact that the count $N(A)$ cannot be negative. The norming axiom reflects the fact that event $\mathcal{S}$ is certain and must occur in every experiment so that $N(\mathcal{S}) = N$, and hence, $N(\mathcal{S})/N = 1$. Finally, (taking the case of two disjoint events A and B), we have $N(A \cup B) = N(A) + N(B)$, since whenever A occurs, B does not, and conversely. Thus, if probability is to reflect the limiting relative frequency, then $P(A \cup B)$ should be equal to $P(A) + P(B)$, since the frequencies satisfy the analogous condition $N(A \cup B)/N = N(A)/N + N(B)/N$.

The need for countable additivity, however, cannot be explained so simply. This need is related to the fact that to build a sufficiently powerful theory, one needs to take limits. If $A_1, A_2, \dots$ is a monotone sequence of events (increasing or decreasing, i.e., $A_1 \subset A_2 \subset \dots$ or $A_1 \supset A_2 \supset \dots$) then $\lim P(A_n) = P(\lim A_n)$, where the event $\lim A_n$ has been defined in Section 1.4. Upon a little reflection, one can see that such continuity is a very natural requirement. In fact, the same requirement has been taken for granted for over 2,000 years in a somewhat different context: in computing the area of a circle, one uses a sequence of polygons with an increasing number of sides, all inscribed in the circle. This leads to an increasing sequence of sets “converging” to the circle, and therefore the area of the circle is taken to be the limit of the areas of approximating polygons. The validity of this idea (i.e., the assumption of the continuity of the function $f(A) = \text{area of } A$) was not really questioned until the beginning of the twentieth century. Research on the subject culminated with the results of Lebesgue.

To quote the relevant theorem, let us say that a function P , defined on a class of sets (events), is *continuous from below at the set A* if the conditions $A_1 \subset A_2 \subset \dots$ and $A = \bigcup A_n$ imply that $\lim P(A_n) = P(A)$. Similarly, P is *continuous from above at the set A* if the conditions $A_1 \supset A_2 \supset \dots$ and $A = \bigcap A_n$ imply $\lim P(A_n) = P(A)$. A function that is continuous

at every set from above or from below is simply called *continuous* (above or below). Continuity from below *and* from above is simply referred to as *continuity*.

We may characterize countable additivity as follows:

Theorem 2.6.1 *If the probability P satisfies Axiom 3 of countable additivity, then P is continuous from above and from below. Conversely, if a function P satisfies Axioms 1 and 2, is finitely additive, and is either continuous from below or continuous from above at the empty set $\emptyset$, then P is countably additive.*

Proof: Assume that P satisfies Axiom 3, and let $A_1 \subset A_2 \subset \cdots$ be a monotone increasing sequence. We have

$$\begin{aligned} \bigcup_{n=1}^{\infty} A_n &= A_1 \cup (A_2 \cap A_1^c) \cup (A_3 \cap A_2^c) \cup \cdots \\ &= A_1 \cup (A_2 \setminus A_1) \cup (A_3 \setminus A_2) \cup \cdots \end{aligned} \quad (2.8)$$

the events on the right-hand side being disjoint. Since $\bigcup A_n = \lim A_n$ (see Section 1.5), using (2.8), and the assumption of countable additivity, we obtain

$$\begin{aligned} P(\lim_{n \rightarrow \infty} A_n) &= P(\bigcup A_n) = P(A_1) + \sum_{i=2}^{\infty} [P(A_i) - P(A_{i-1})] \\ &= P(A_1) + \lim_{n \rightarrow \infty} \sum_{i=2}^n [P(A_i) - P(A_{i-1})] \\ &= P(A_1) + \lim_{n \rightarrow \infty} [P(A_n) - P(A_1)] = \lim_{n \rightarrow \infty} P(A_n) \end{aligned}$$

(passing from the first to the second line, we used the fact that the infinite series is defined as the limit of its partial sums). This proves continuity of P from below. To prove continuity from above, we pass to the complements, and proceed as above.

Let us now assume that P is finitely additive and continuous from below, and let $A_1, A_2, \dots$ be a sequence of mutually disjoint events. Put $B_n = A_1 \cup \cdots \cup A_n$ so that $B_1 \subset B_2 \subset \cdots$ is a monotone increasing sequence with $\bigcup A_n = \bigcup B_n$. We have then, using continuity from below and finite additivity,

$$\begin{aligned} P(\bigcup A_n) &= P(\bigcup B_n) = P(\lim_{n \rightarrow \infty} B_n) = \lim_{n \rightarrow \infty} P(B_n) \\ &= \lim_{n \rightarrow \infty} [P(A_1) + \cdots + P(A_n)] = \sum_{n=1}^{\infty} P(A_n). \end{aligned}$$

again by definition of a numerical series being the limit of its partial sums. This shows that P is countably additive.

Finally, let us assume that P is finitely additive and continuous from above at the empty set $\emptyset$ (impossible event). Taking again a sequence of disjoint events $A_1, A_2, \dots$, let $C_n = A_{n+1} \cup A_{n+2} \cup \cdots$. We have $C_1 \supset C_2 \supset \cdots$ and $\lim C_n = \bigcap_n C_n = \emptyset$. By finite additivity, we obtain

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = P\left(\bigcup_{i=1}^n A_i \cup C_n\right) = P(A_1) + \cdots + P(A_n) + P(C_n). \quad (2.9)$$

Since (2.9) holds for every n , we can write

$$\begin{aligned} P(\cup A_i) &= \lim[P(A_1) + \cdots + P(A_n) + P(C_n)] \\ &= \lim[P(A_1) + \cdots + P(A_n)] + \lim P(C_n) = \sum_{n=1}^{\infty} P(A_i). \end{aligned}$$

Again, by the definition of series and the assumption that $\lim P(C_n) = 0$, P is countably additive, and the proof is complete. $\square$

As an illustration, we now prove the following theorem:

Theorem 2.6.2 (First Borel–Cantelli Lemma) *If $A_1, A_2, \dots$ is a sequence of events such that*

$$\sum_{n=1}^{\infty} P(A_n) < \infty, \quad (2.10)$$

then

$$P(\limsup A_n) = 0.$$

Proof: Recall (1.7) from Chapter 1, where $\limsup A_n =$ “infinitely many events A_j occur” $= \bigcap_{k=1}^{\infty} \bigcup_{i=k}^{\infty} A_i = \lim_{k \rightarrow \infty} \bigcup_{i=k}^{\infty} A_i$ (because the unions $\bigcup_{i=k}^{\infty} A_i, k = 1, 2, \dots$ form a decreasing sequence). Consequently, using the continuity of P , subadditivity property (2.2), and assumption (2.10), we have

$$P(\limsup A_n) = \lim_{k \rightarrow \infty} P\left(\bigcup_{i=k}^{\infty} A_i\right) \leq \lim_{k \rightarrow \infty} \sum_{i=k}^{\infty} P(A_i) = 0. \quad \square$$

Paraphrasing the assertion of the lemma, if probabilities of events A_j decrease to zero fast enough to make the series converge, then with probability 1 only finitely many among events A_j will occur. We will prove the converse (under an additional assumption), known as the second Borel–Cantelli lemma, in Chapter 4.

In the remainder of this section, we will discuss some theoretical issues related to defining probability in practical situations. Let us start with the observation that the analysis above should leave some more perceptive readers disturbed. Clearly, one should not write a formula without being certain that it is well defined. In particular, when writing $P(\cdots)$ two things ought to be certain: (1) that what appears in the parentheses is a legitimate object of probability, that is, an event and (2) that the function P is defined unambiguously at this event.

With regard to the first point, the situation is rather simple. All reasonable questions concern events such as $\lim A_n$ and $\limsup A_n$, and hence events obtained by taking countable unions, countable intersections, and complementations of the events $A_1, A_2, \dots$. Thus, the events whose probabilities are discussed belong to the smallest σ -field containing all the events $A_1, A_2, \dots$ (see Definition 1.4.2 and Theorem 1.4.3). Consequently, to make the formulas at least apparently legitimate, it is sufficient to assume that the class of all the events under considerations is a σ -field, and that probability is a function satisfying the probability axioms defined on this σ -field.

This assumption alone, however, is not enough to safeguard us from possible trouble. Let us consider the following hypothetical situation: Suppose that we do not know how to calculate the area of a circle. We could start from the beginning and define the areas of simple figures: first rectangles, then pass to right triangle, and then to arbitrary triangles,

which could be reduced to sums and differences of right triangles. From there, the concept of area could be extended to figures that could be triangulated. It is a simple matter to show that the area of such a figure does not depend on how it is triangulated.

From here, we may pass to areas of more complicated figures, the first of these being the circle. The area of the latter could be calculated by inscribing a square in it, and then taking areas of regular polygons with 8, 16, 32, . . . sides and passing to the limit. The result is πr^2 . The same result is obtained if we start by inscribing an equilateral triangle, and then take limits of the areas of regular polygons with 6, 12, 24, . . . sides. The same procedure could be tried with an approximation from above, that is, starting with a square or equilateral triangle circumscribed on the circle. Still the limit is πr^2 . We could then be tempted to conclude that the area of the circle is πr^2 . The result is, of course, true, but how do we know that we will obtain the limit *always* equal to πr^2 , regardless of the way of approximating the circle? What if we start, say, from an irregular seven-sided polygon, and then triple the number of sides in each step?

A similar situation occurs very often in probability: Typically, we can define probabilities on “simple” events, corresponding to rectangles in geometry, and we can extend this definition without ambiguity to finite unions of the simple events (“rectangles”). The existence and uniqueness of a probability of all the events from the minimal σ -field containing the “rectangles” is ensured by the following theorem, which we state here without proof.

Theorem 2.6.3 *If P is a function defined on a field of events $\mathcal{A}$ satisfying the probability axioms (including countable additivity), then P can be extended in a unique way to a function satisfying the probability axioms, defined on the minimal σ -field containing $\mathcal{A}$.*

This means that if the function P is defined on a field $\mathcal{A}$ of events and satisfies all the axioms of probability, and if $\sigma(\mathcal{A})$ is the smallest σ -field containing all sets in $\mathcal{A}$, then there exists exactly one function P^* defined on $\sigma(\mathcal{A})$ that satisfies the probability axioms, and $P^*(A) = P(A)$ if $A \in \mathcal{A}$.

A comment that is necessary here concerns the question: What does it mean that a function P defined on a field $\mathcal{A}$ satisfies the axioms of probability? Specifically, the problem concerns the axiom of countable additivity, which asserts that if events $A_1, A_2, \dots$ are disjoint, then

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n). \quad (2.11)$$

However, if P is defined on a field, then there is no guarantee that the left-hand side of formula (2.11) makes sense, since $\bigcup_{n=1}^{\infty} A_n$ need not belong to the field of events on which P is defined. The meaning of the assumption of Theorem 2.6.3 is that formula (2.11) is true whenever the union $\bigcup_{n=1}^{\infty} A_n$ belongs to the field on which P is defined.

The way of finding the probability of some complicated event A is to represent A as a limit of some sequence of events whose probabilities can be computed, and then pass to the limit. Theorem 2.6.3 asserts that this procedure will give the same result, regardless of the choice of sequence of events approximating the event A .

■ EXAMPLE 2.7 Densities

A very common situation in probability theory occurs when $\mathcal{S} = (-\infty, +\infty)$. A probability measure P on $\mathcal{S}$ can be defined as follows: let f be a function such that $f(x) \geq 0$ for all x and $\int_{-\infty}^{+\infty} f(x) dx = 1$. We will assume in addition that f is continuous and bounded, although those conditions can be greatly relaxed in general theory.

We now *define* probability on $\mathcal{S}$ by putting

$$P(A) = \int_A f(x) dx \quad (2.12)$$

(in this case, f is referred to as a *density* of P). The full justification of this construction lies beyond the scope of this book, but we will give the main points. First, the definition (2.12) is applicable for all intervals A of the form (a, b) , $[a, b]$, $(-\infty, b)$, (a, ∞) , $[a, \infty)$, and so on. Then we can extend P to finite unions of disjoint intervals by additivity (the class of all such finite unions forms a field). We can easily check that such an extension is unique; that is,

$$P[(a, b)] = \int_a^b f(x) dx = \sum_j \int_{I_j} f(x) dx$$

does not depend on the way interval (a, b) is partitioned into the finite union of nonoverlapping intervals I_j . This provides an extension of P to the smallest field of sets containing all intervals. If we show that P defined this way is continuous on the empty set, then we can claim that there exists an extension of P to the smallest σ -field of sets containing all intervals.

Now, the decreasing sequences of intervals converging to the empty set are built of two kinds of sequences: “shrinking open sets” and “escaping sets,” exemplified as

$$I_1 \supset I_2 \supset \cdots \quad \text{with} \quad I_n = (a, a + \epsilon_n), \epsilon_1 > \epsilon_2 > \cdots \rightarrow 0$$

and

$$J_1 \supset J_2 \supset \cdots \quad \text{with} \quad J_n = (a_n, \infty), a_1 < a_2 < \cdots \rightarrow \infty.$$

We have here $\lim I_n = \bigcap I_n = \emptyset$ and $\lim J_n = \bigcap J_n = \emptyset$. In the first case, $P(I_n) = \int_a^{a+\epsilon_n} f(x) dx \leq \epsilon_n M \rightarrow 0$, where M is a bound for function f . In the second case, $P(J_n) = \int_{a_n}^{\infty} f(x) dx = 1 - \int_{-\infty}^{a_n} f(x) dx \rightarrow 1 - \int_{-\infty}^{+\infty} f(x) dx = 0$.

2.7 SUBJECTIVE PROBABILITY*

Let us finally consider briefly the third interpretation of probability, namely as a degree of certainty, or belief, about the occurrence of an event. Most often, this probability is associated not so much with an event as with the truth of a proposition asserting the occurrence of this event.

The material of this section assumes some degree of familiarity with the concept of expectation, formally defined only in later chapters. For the sake of completeness, in the simple form needed here, this concept is defined below. In the presentation, we follow more or less the historical development, refining gradually the conceptual structures introduced. The basic concept here is that of a *lottery*, defined by an event, say A , and two objects, say a and b . Such a lottery, written simply aAb , will mean that the participant (X) in the lottery receives object a if the event A occurs, and receives object b if the event A^c occurs.

The second concept is that of *expectation* associated with the lottery aAb , defined as

$$u(a)P(A) + u(b)P(A^c), \quad (2.13)$$

where $u(a)$ and $u(b)$ are measures of how much the objects a and b are “worth” to the participant. When a and b are sums of money (or prices of objects a and b), and we put $u(x) = x$, the quantity (2.13) is sometimes called *expected value*. In cases where $u(a)$ and $u(b)$ are values that person X attaches to a and b (at a given moment), these values do not necessarily

coincide with prices. We then refer to $u(a)$ and $u(b)$ as *utilities* of a and b , and the quantity (2.13) is called *expected utility* (EU). Finally, when in the latter case, the probability $P(A)$ is the subjective assessment of likelihood of the event A by X , the quantity (2.13) is called *subjective expected utility* (SEU).

First, it has been shown by Ramsey (1926) that the degree of certainty about the occurrence of an event (of a given person) can be measured. Consider an event A , and the following choice suggested to X (whose subjective probability we want to determine). X is namely given a choice between the following two options:

1. *Sure* option: receive some fixed amount $\$u$, which is the same as lottery $(\$u)B(\$u)$, for any event B .
2. A *lottery* option. Receive some fixed amount, say $\$100$, if A occurs, and receive nothing if A does not occur, which is lottery $(\$100)A(\$0)$. One should expect that if u is very small, X will probably prefer the lottery. On the other hand, if u is close to $\$100$, X may prefer the sure option.

Therefore, there should exist an amount u^* such that X will be indifferent between the sure option with u^* and the lottery option. With the amount of money as a representation of its value (or utility), the expected return from the lottery equals

$$0(1 - P(A)) + 100P(A) = 100P(A),$$

which, in turn, equals u^* . Consequently, we have $P(A) = u^*/100$. Obviously, under the stated assumption that utility of money is proportional to the dollar amount, the choice of $\$100$ is not relevant here, and the same value for $P(A)$ would be obtained if we choose another “base value” in the lottery option (this can be tested empirically).

This scheme of measurement may provide an assessment of the values of the (subjective) probabilities of a given person, for a class of events. It is of considerable interest that the same scheme was suggested in 1944 by von Neumann and Morgenstern (1944) as a tool for measuring utilities. They assumed that probabilities are known (i.e., the person whose utility is being assessed knows the objective probabilities of events, and his subjective and objective probabilities coincide). If a person is now indifferent between the lottery as above, and the sure option of receiving an object, say q , then the utility $u(q)$ of object q must equal the expected value of the lottery, which is $100P(A)$. This allows one to measure utilities on the scale that has a zero set on nothing (status quo) and “unit” as the utility of $\$100$. The scheme of von Neumann and Morgenstern was later improved by some authors, culminating with the theorem of Blackwell and Girshick (1954).

Still the disadvantages of both approaches were due to the fact that to determine utilities, one needed to assume knowledge of probabilities by the subject, while conversely, to determine subjective probabilities, one needed to assume knowledge of utilities. The discovery that one can determine *both* utilities and subjective probabilities of the same person is due to Savage (1954). We present here the basic idea of the experiment rather than formal axioms (to avoid obscuring the issue by technicalities).

Let $A, B, C, \dots$ denote events, and let $a, b, c, \dots$ denote some objects, whose probabilities $P(A), P(B), \dots$ and utilities $u(a), u(b), \dots$ are to be determined (keep in mind that both P and u refer to a particular person X , the subject of the experiment). We now accept the main postulate of the theory, that of the two lotteries, X will prefer the one that has higher SEU .

Suppose that we find an event A with subjective probability $1/2$, so that $P(A) = P(A^c) = 1/2$. If X prefers lottery aAb to lottery cAd , then

$$u(a)P(A) + u(b)P(A^c) > u(c)P(A) + u(d)P(A^c),$$

which means that

$$u(a) - u(c) > u(d) - u(b).$$

A number of experiments on selected objects will allow us to estimate the utilities, potentially with an arbitrary accuracy (taking two particular objects as zero and a unit of the utility scale). In turn, if we know the utilities, we can determine the subjective probability of any event B . That is, if X is indifferent between lotteries aBb and cBd , we have

$$u(a)P(B) + u(b)(1 - P(B)) = u(c)P(B) + u(d)(1 - P(B)),$$

which gives

$$P(B) = \frac{u(d) - u(b)}{u(a) - u(b) + u(d) - u(c)}.$$

The only problem lies in finding an event A with subjective probability $1/2$. Empirically, an event A has subjective probability $1/2$ if, for any objects a and b , the person is indifferent between lotteries aAb and bAa . Such an event was found experimentally (Davidson et al., 1957). It is related to a toss of a die with three of the faces marked with the nonsense combination ZOJ , and the other three with the nonsense combination ZEJ (these combinations evoked the least number of associations).

Let us remark at this point that the system of Savage involves determining first an event with probability $1/2$, then the utilities, and then the subjective probabilities. Luce and Krantz (1971) suggested an axiom system (leading to an appropriate scheme) that allows *simultaneous* determination of utilities and probabilities. The reader interested in these topics is referred to the monograph by Krantz et al. (1971).

A natural question arises: Are the three axioms of probability theory satisfied here (at least in their finite versions, without countable additivity)? On the one hand, this is the empirical question: The probabilities of various events can be determined numerically (for a given person), and then used to check whether the axioms hold. On the other hand, a superficial glance could lead one to conclude that there is no reason why person X 's probabilities should obey any axioms: After all, subjective probabilities that do not satisfy probability axioms are not logically inconsistent.

However, there is a reason why a person's subjective probabilities should satisfy the axioms. For any axiom violated by the subjective probability of X (and X accepts the principle of SEU), one could design a bet that appears favorable to X (hence a bet that he will accept), but yet the bet is such that X is sure to lose.

Indeed, suppose first that the probability of some event A is negative. Consider the bet (lottery) $(-c)A(-b)$, (i.e., a lottery in which X pays the sum c if A occurs and pays the sum b if A does not occur). We have here (identifying, for simplicity, the amounts of money with their utilities)

$$\text{SEU} = -cP(A) - bP(A^c),$$

so that SEU is positive for a large enough c if $P(A) < 0$. Thus, following the principle of maximizing SEU, X should accept this lottery over the status quo (no bet), but he will lose in any case—the amount c or the amount b .

Suppose now that $P(S) < 1$. Consider the bet $(-c)Sb$ whose SEU is $-cP(S) + bP(S^c)$. Since $P(S^c) > 0$, making b large enough, the bet appears favorable to X , yet he is bound to lose the amount c on every trial.

If $P(S) > 1$ or if the additivity axiom is not satisfied, one can also design bets that will formally be favorable for X (SEU will be positive) but that X will be bound to lose. Determination of these bets is left to the reader.

PROBLEMS

- 2.7.1** Peter and Tom attend the same college. One day Tom buys a ticket for a rock concert. Tickets are already sold out and are in great demand. Peter, who does not have a ticket, agrees to play the following game with Tom. For a fee of \$25, Peter will toss a coin three times and receive the ticket if all tosses show up heads. Otherwise, for an additional fee of \$50, Peter will toss a coin two more times and receive the ticket if both tosses show up heads. If not, then for an additional fee of \$100, Peter will toss a coin and receive the ticket if the toss shows up heads. Otherwise, all money will be given to Tom, and he also will keep the ticket. Assuming that the coin is fair, subjective probabilities of various outcomes coincide with objective probabilities, and that Peter's utility is linear in money, show that Peter's utility of the ticket exceeds \$200.
- 2.7.2** Refer to Problem 2.7.1. Tom would agree on the following conditions: Peter pays him \$50 and tosses a coin, winning the ticket if it comes up heads, and otherwise losing \$50. In such a situation, should they both agree that Peter buys the ticket from Tom for \$150?
- 2.7.3** Suppose that Tom is confronted with the choice between two options: O_1 , which is simply to receive \$1,000,000, or O_2 , which is to receive \$5,000,000 with probability 0.1, receive \$1,000,000 with probability 0.89, and receive \$0 with the remaining probability 0.01. After some deliberation, Tom decides that O_1 is better, mostly because the outcome \$0, unlikely as it may be, is very unattractive. Tom is also confronted with a choice between two other options, O_3 and O_4 . In O_3 , he would receive \$5,000,000 with probability 0.1 and \$0 with probability 0.9. In O_4 , he would receive \$1,000,000 with probability 0.11 and \$0 with probability 0.89. Here Tom prefers O_3 : the "unattractive" option \$0 has about the same probability in both O_3 and O_4 , while the positive outcome, although slightly less probable under O_3 , is much more desirable in O_3 than in O_4 . Show that these preferences of Tom are not compatible with the assumption that he has utilities A , B , and C of \$5,000,000, \$1,000,000, and \$0, such that $A > B > C$ (This is known as Allais' paradox; Allais, 1953).